

Research Paper

Enhanced flow uniformity in parallel mini-channels with pin-finned inlet header

Jung-Youn Song^a, Seungryong Hah^b, Deokjoo Kim^b, Sung-Min Kim^{a,*}^a School of Mechanical Engineering, Sungkyunkwan University, 300 Cheoncheon-dong, Suwon 16419, Republic of Korea^b Agency for Defense Development, P.O. Box 35, Daejeon 34186, Republic of Korea

HIGHLIGHTS

- A new strategy utilizing pin-finned inlet header to improve flow uniformity in parallel mini-channels.
- Flow patterns in inlet header such as transverse and circulating flows are identified.
- The best pin fin arrangements in both trapezoidal and rectangular inlet headers are obtained.
- The use of pin fins in inlet header has minimal influence on pressure drop.
- A noticeable improvement in heat-sink thermal performance is realized for best layouts.

ARTICLE INFO

Keywords:

Heat sink

Staggered pin-fin array

Flow maldistribution

Pressure drop

ABSTRACT

A key challenge concerning the design of mini-channel heat sinks pertains to minimization of flow maldistribution in parallel channels. Although this challenge has spurred several research efforts towards exercising control over flow resistance within parallel channels to enhance flow uniformity, some practical issues associated with fabrication and implementation, as well as the possibility of a high pressure drop render the efforts less useful. The present study concerns development of a strategy that utilizes staggered pin-fin arrays in the inlet header of mini-channel heat sinks with the aim of achieving uniform flow distribution within parallel channels without increasing total pressure drop. Two different header shapes, trapezoidal and rectangular, have been considered, and ten different pin-fin arrangements for each type of header have been designed to numerically investigate flow uniformity within channels. The three-dimensional computational domain included the entire heat sink consisting of inlet/outlet connection tubes, inlet/outlet headers, and twenty parallel rectangular mini-channels—each measuring 3 mm in hydraulic diameter and equidistantly spaced with a 2-mm solid-wall spacing. It has been demonstrated that compared to baseline inlet headers, best pin-fin arrangements obtained for both trapezoidal and rectangular inlet headers yield significant improvement in flow uniformity within channels along with uniform temperature distribution over the heat sink base with minimal influence on the pumping power requirement.

1. Introduction

The ability to remove high heat flux from a small surface area is a common requirement in many modern applications, such as avionics, hybrid/electric-vehicle power electronics, storage systems for hydrogen fuel cells, computer data centers, defense lasers, and microwave weapon systems. As heat dissipation from devices used in above applications continues to escalate, air-cooling schemes are no longer capable of coping with the increased cooling demands [1]. As an

effective alternative, various liquid-cooling schemes, including mini/micro-channel flow, jet-impingement, spray, and direct immersion, have been explored [1]. Among these cooling schemes, the use of mini/micro-channel heat sinks has received remarkable attention owing to it offering a number of advantages, such as compactness, small coolant inventory requirement, and high heat-dissipation-to-volume ratio [1,2].

These advantages have spurred numerous research efforts aimed at understanding various aspects of two-phase flow and heat transfer mechanisms, specifically tailored to mini/micro-channel flows [3–9].

* Corresponding author.

E-mail address: smkim@skku.edu (S.-M. Kim).URL: <http://htl.skku.edu> (S.-M. Kim).<https://doi.org/10.1016/j.applthermaleng.2019.02.069>

Received 20 July 2018; Received in revised form 11 February 2019; Accepted 15 February 2019

Available online 16 February 2019

1359-4311/ © 2019 Elsevier Ltd. All rights reserved.

Nomenclature

A_b	heat sink base area
A_{ch}	cross-sectional area of mini-channel
C_1, C_2	empirical constants in turbulent dissipation rate equation
c_p	constant pressure specific heat
D_h	channel hydraulic diameter
D_{in}	diameter of inlet connection tube
f	friction factor
G_k	turbulent energy production
I	turbulence intensity
k	turbulent kinetic energy; thermal conductivity
L_{ch}	channel length
MF	maldistribution factor
n	number of channels
P	pressure
Pr_t	turbulent Prandtl number
ΔP_{ch}	channel pressure drop
ΔP_{tot}	total pressure drop between inlet and outlet connection tubes
Q_{avg}	mass flow rate averaged over all mini-channels
Q_{in}	mass flow rate in inlet connection tube
Q_k	mass flow rate in kth channel
q''	heat flux applied at heat sink base
Re	channel Reynolds number
Re_{avg}	channel Reynolds number averaged over all mini-channels
Re_{in}	Reynolds number within inlet connection tube
R_{th}	overall thermal resistance of heat sink
S	modulus of mean rate-of-strain tensor
T	temperature
$T_{b,max}$	maximum temperature of heat sink base

$T_{b,min}$	minimum temperature of heat sink base
ΔT_b	maximum temperature difference of heat sink base
U	time-averaged velocity
$\bar{U}$	average velocity within channel
x, y, z	spatial coordinates
Y_k	turbulent energy dissipation

Greek symbols

β	channel aspect ratio
ε	turbulent dissipation rate
η	strain rate
μ	dynamic viscosity
μ_t	turbulent viscosity
ν	kinematic viscosity
ρ	density
σ_ε	empirical constant in turbulent dissipation rate equation
σ_k	empirical constant in turbulent kinetic energy equation

Subscripts

app	apparent
b	heat sink base
d	fully-developed
f	fluid
in	inlet connection tube
i, j	direction indices
max	maximum
min	minimum
s	solid

Several studies have focused on offering practical flow pattern maps or flow pattern transition criteria based on their observation of two-phase flow regime in mini/micro-channels [3–5]. A few researchers performed flow boiling and condensation experiments under reduced gravity conditions to examine influences of different gravitational conditions on two-phase flow physics and heat transfer [6,7]. Another topic that has recently attracted increased research attention pertains to characterization of two-phase flow boiling instabilities via examination of transient behaviors of different heat sink design parameters, such as pressure drop, mass flux, and thermodynamic equilibrium quality, thereby investigating conditions under which instability modes are manifested [8,9]. Although aforementioned studies provided valuable insight into several aspects of two-phase transport behaviors in mini/micro-channels, availability of information pertaining to practical heat sink design remains rather limited. This limitation has served as the primary motivation for research efforts undertaken by the present authors to gain better understanding of two-phase flow and heat transfer characteristics in mini/micro-channels for the purpose of achieving reliable design and safe operation of mini/micro-channel heat sinks [10–18]. To develop a mechanistic understanding of annular two-phase flows in mini/micro-channels, theoretical control-volume-based models for annular flow boiling [10] and annular flow condensation [11] have been constructed. In pursuit of a systematic and comprehensive methodology for thermal design of mini/micro-channel heat sinks, generalized predictive tools have been developed using large databases that encompass very broad ranges of operating parameters and many working fluids. These include two-phase pressure drop for boiling [12] as well as condensing/adiabatic [13] flows, two-phase heat transfer coefficient for boiling [14] and condensing [15] flow, and dryout incipience quality [16]. Thermal-design limitations associated with premature critical heat flux (CHF), dryout incipience, and two-phase critical flow [17] in micro-channel heat sinks, as well as relationships

between CHF, premature CHF, and two-phase critical flow in micro-channel heat sinks [18] have been examined.

One of the key challenges involved in the design of mini/micro-channel heat sinks corresponds to the attainment of uniform flow distribution within parallel mini/micro-channels. Flow maldistribution within parallel channels contributes to flow-induced vibrations as well as non-uniform temperature distribution within heat sinks [19]. This, in turn, results in mechanical and thermal stresses being induced within heat sinks, thereby posing significant risks, such as deformation or warping of the heat sink material. In particular, as regards the thermal design of heat sinks undergoing phase change along channels intended for high-heat-flux cooling, flow maldistribution may cause dryout incipience in some channels, in cases wherein local portions of the annular liquid film begin to dry out, as well as critical heat flux (CHF) triggered by complete dryout of the annular liquid film, thereby resulting in catastrophic failure of heat sinks due to overheating [18,20].

Flow distribution in mini/micro-channel heat sinks can be greatly influenced by their geometrical parameters. A mini/micro-channel heat sink consists of multiple parallel channels, inlet/outlet headers, and inlet/outlet connection tubes. The inlet header plays the role of distributing the flow into multiple parallel channels, whereas the outlet header is responsible for collecting flow from the channels and guiding to the outlet connection tube. Extensive research efforts have been directed towards reduction of flow maldistribution via optimization of geometrical parameters concerning inlet/outlet headers, such as header shape, its orientation, and inlet/outlet flow arrangement. Chein and Chen [21] investigated the effect of different inlet/outlet arrangements (I-, D-, N-, S-, U- and V-types) on flow uniformity within micro-channel heat sinks. Inlet/outlet arrangements used in their study were classified into the horizontal inlet/outlet arrangement, wherein the fluid enters and leaves the inlet and outlet headers horizontally (i.e., I-, D-, N- and S-types), and the vertical inlet/outlet arrangement, which has a vertical

fluid supply and collection (i.e., U- and V-types). Results obtained demonstrated that the horizontal inlet/outlet arrangements provide poor flow uniformity within channels due to the existence of flow recirculation at corners of inlet/outlet headers, whereas in vertical inlet/outlet arrangements, flow impingement in the inlet header yields appreciable enhancement in flow uniformity. Lu and Wang [22] performed numerical simulations to examine flow maldistribution within mini-channel heat sinks using different inlet/outlet configurations (I-, Z-, J-, L- and T-types). Their results revealed that a comparatively uniform flow distribution is obtained from the J-type configuration by reducing flow recirculation. The I-type configuration with vertical inlet/outlet arrangement demonstrated the highest pressure drop, whereas the J-type demonstrated the lowest. Kumaran et al. [23] performed experiments and numerical simulations to examine the effect of different inlet/outlet header shapes (rectangular, trapezoidal and triangular), header sizes, and different inlet/outlet arrangements (I-, C-, V-, Z-, and U-types) on flow distribution. Their result demonstrated that for the inlet header, the triangular shape is advantageous compared to other shapes, whereas the trapezoidal shape is superior to other shapes with regard to the outlet header. They further reported that the C-type configuration—or one having a smaller header width—show better flow uniformity within channels. They concluded that flow maldistribution is considerably influenced by flow recirculation and separation effects within the inlet header; this is consistent with the result reported by Chein and Chen [21]. Numerical simulations were performed by Tong et al. [24] to assess the influence of geometrical parameters, such as header-height-to-channel-width ratio, tapering shape of the header, and cross-sectional area of inlet header, on flow distribution in heat sinks. Better flow distribution was achieved through use of a larger header height and greater cross-sectional area of the inlet header. Siva et al. [25] performed experiments to explore the effect of three different inlet/outlet arrangements (I-, Z-, and U-types) on flow distribution within channels. Results revealed that the I-type provides the smallest total pressure drop along with best flow uniformity within channels. Mishan et al. [26] performed numerical investigations to compare the effect of three different header designs (rectangular, straight-convergent, and divergent-straight-convergent shape) on flow distribution. Results revealed that the conventional rectangular header provides poor flow uniformity within channels, whereas the uniformity is fairly enhanced when employing the straight-convergent header. An experimental study was performed by Anbumeenkshi and Thensekhar [27] to examine the effects of inlet arrangements (horizontal and vertical), inlet header shapes (rectangular, trapezoidal and triangular), and total flow rates on flow distribution within channels. The results showed that relatively better flow uniformity in channels is obtained when employing the vertical inlet arrangement due to the flow impingement effect; this is in agreement with numerical simulation results reported by Chein and Chen [21]. Another noteworthy observation is that the rectangular header provides poor flow uniformity at a lower inlet flow rate, whereas the most uniform flow distribution is obtained at a higher inlet flow rate. Numerical simulations were performed by Xia et al. [28] to examine the effects of different inlet/outlet header shapes (rectangular, symmetric trapezoidal, and symmetric triangular), and inlet/outlet arrangements (C-, Z- and I-type) on flow distribution. Their results showed that the rectangular header is superior to the other header shapes, and that flow distribution within channels remains comparatively uniform when employing the I-type inlet/outlet arrangement while the C- and Z-types exhibit poor flow distribution within channels due to intense flow separation in the inlet header.

There have been several attempts of controlling flow resistance within parallel channels to improve flow uniformity by means of installing baffles, orifices, and converging nozzles either in the entrance region of channels or within the inlet header. Wang et al. [29] experimentally examined flow distribution characteristics by modifying the inlet header with baffle plates and tubes. They tested 2 different baffle plates and 7 baffle tubes possessing multiple evenly distributed

circular holes of different sizes. Their modified inlet header with baffle tube demonstrated the best flow distribution compared to others by suppressing vortex flows inside the inlet header. Liu and Yu [30] numerically investigated the effect of nonuniform mini baffles placed at the channel inlet of mini-channel heat sinks. They showed that the use of mini baffles of different widths for each channel yields significant reduction in the overall thermal resistance of heat sinks due to enhanced flow uniformity in channels. Despite research efforts to reduce flow maldistribution using above-mentioned methods, some practical issues associated with fabrication and implementation, as well as the possibility of high pressure drop render them less useful.

While flow maldistribution is a common concern in all types of mini/micro-channel heat sinks, it is especially problematic for mini-channel heat sinks with relatively large channel diameters. In such heat sinks, the pressure drop in channels is a small fraction of the total pressure drop between the inlet and outlet connection tubes; therefore, the flow rate in each channel is dictated mainly by the contraction pressure loss and expansion pressure recovery at the inlet and outlet of the channel, respectively. In micro-channel heat sinks, on the other hand, flow maldistribution is less influenced by geometrical parameters of inlet/outlet headers due to the large pressure drop in channels, as discussed in [22,31,32]. Accordingly, the present study concerns mini-channel heat sinks utilizing staggered pin-fin arrays inside the inlet header with the aim of achieving uniform flow distribution within parallel channels without increasing the total pressure drop. Two different header shapes, trapezoidal and rectangular, have been considered along with ten different pin-fin arrangements for each type of header designed to examine flow uniformity within channels.

2. Numerical modeling

2.1. Model descriptions

The effect of pin-fin arrangements in the inlet header of mini-channel heat sinks on the flow distribution and total pressure drop in parallel channels has been numerically investigated in this study. The three-dimensional computational domain considered for this purpose includes the entire heat sink comprising multiple parallel mini-channels, inlet and outlet headers, and inlet and outlet connection tubes, as depicted in Fig. 1. To demonstrate the effectiveness of installing pin fins for minimizing flow maldistribution, the V-type inlet/outlet arrangement has been used owing to it being known to provide relatively poor flow uniformity within channels compared to other types of inlet/outlet arrangement [23,28]. Two different header shapes, trapezoidal and rectangular, which have been widely employed in numerous manifold systems [23,27–32], have been taken into consideration. A heat sink base area, A_b , measuring $80 \times 100 \text{ mm}^2$ was chosen to encompass a range of heat sink base areas associated with the cooling of electronic devices, such as insulated-gate bipolar transistor (IGBT) power modules, and computer CPUs. Twenty parallel channels were equally spaced by means of a 2-mm solid-wall spacing between adjacent channels. Each channel measured 2-mm wide and 6-mm deep, thereby yielding a hydraulic diameter of 3 mm, which corresponds to the upper limit of mini-channel diameter according to a threshold channel diameter of 3 mm between mini-channels and macro-channels, as suggested by Kandlikar and Grande [33]. Since mini-channel heat sinks generally operate in the laminar flow regime due to the practical limitations associated with pumping power [34], three different inlet mass flow rates Q_{in} of 34.1, 68.3, and 102.4 g/s, corresponding to Re_{avg} values of 427, 853, and 1280, respectively, were selected for the numerical simulations. The channel Reynolds number averaged over all mini-channels at a constant temperature of 20 °C, Re_{avg} , could be expressed as

$$Re_{avg} = \frac{Q_{in} D_{in}}{\mu A_{ch} n}, \quad (1)$$

where D_{in} , A_{ch} , and n denote the inlet-connection-tube diameter, mini-

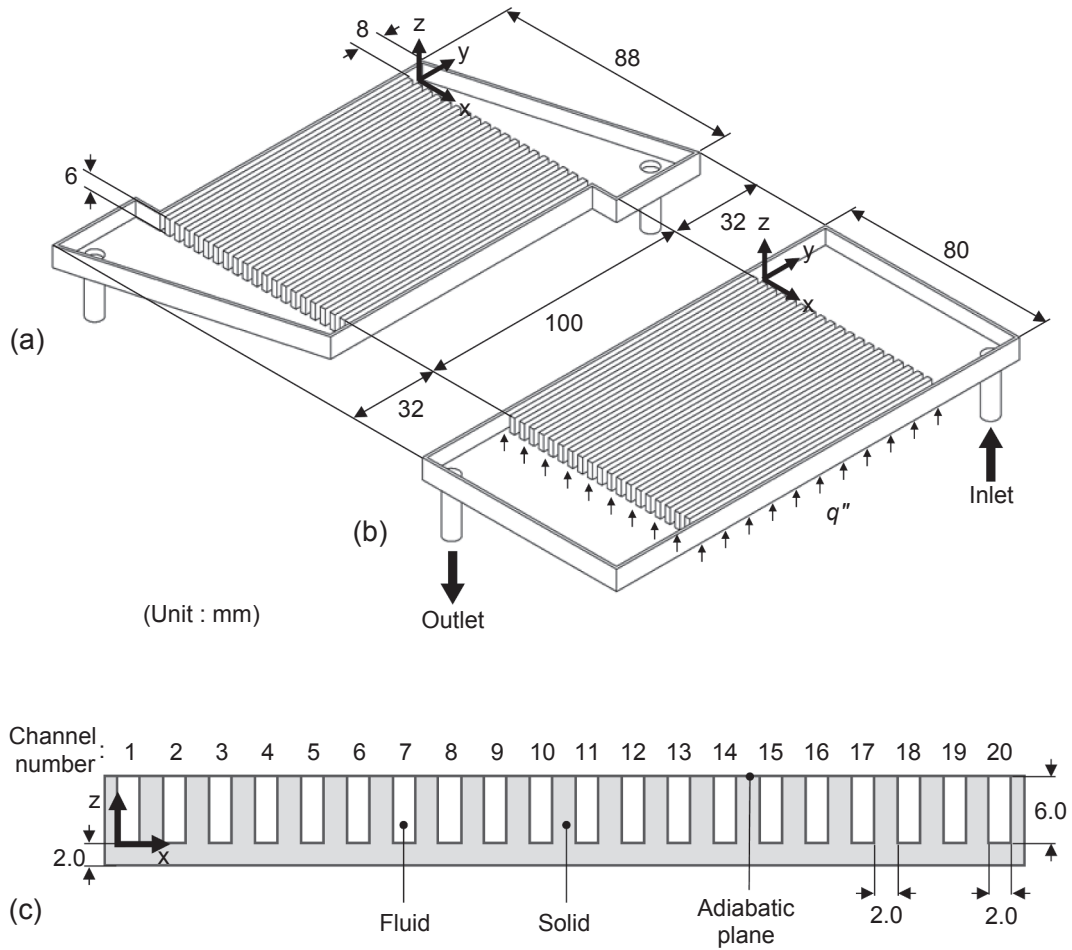


Fig. 1. Mini-channel heat sinks with (a) trapezoidal and (b) rectangular headers. (c) Cross-sectional view of parallel mini-channels.

channel cross-sectional area, and number of channels, respectively.

Staggered pin-fin arrays in both the trapezoidal and rectangular inlet headers comprised three rows positioned along the y-direction, as illustrated in Fig. 2(a) and (b), respectively. Identical pin fins with a diameter of 2.8 mm and a height of 6 mm were used for both inlet headers. The trapezoidal inlet header could be occupied by up to 57 pin fins with longitudinal- and transverse-pitch values of 5.5 mm and 3.6 mm, respectively. Given the geometrical discrepancy between the two headers, the rectangular inlet header allowed a maximum of 62 pin fins with longitudinal- and transverse-pitch values of 5.5 mm and 3.7 mm, respectively.

2.2. Numerical method

Ansys Fluent 18.0 [35], the commercial CFD software, was used for performing numerical simulations in this study. Owing to high flow velocities in the vicinity of pin fins as well as flow maldistribution within parallel channels, there exists a possibility of the flow to become locally turbulent, despite existence of relatively small Re_{avg} values (i.e., 427, 853 and 1280) of less than 2000, thereby indicating laminar flow in channels in the case of perfect uniform distribution. In addition, the value of the Reynolds number, Re_{in} , within the inlet connection tube measured in the range of 7224–21,695, thus indicating existence of turbulent flow inside the tube, albeit Re_{avg} values remained relatively low. For these reasons, the realizable $k-\epsilon$ turbulence model with enhanced wall treatment [35,36] was employed owing to it being known to produce reliable predictions for planar as well as round jets, rotating flows, and boundary layers under strong adverse pressure gradients and recirculation [35]. Additionally, the semi-implicit method for pressure-

linked equations-consistent (SIMPLEC) algorithm [37] was employed to obtain pressure-velocity coupling. Second-order and second-order upwind schemes are used for pressure and momentum discretization, respectively.

The time-averaged governing equations for steady-state, incompressible fluid flow could be expressed as

Continuity equation:

$$\frac{\partial}{\partial x_j}(\rho U_j) = 0 \quad (2)$$

Momentum equation:

$$\frac{\partial}{\partial x_j}(\rho U_i U_j) = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j} \left[(\mu + \mu_t) \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right) \right] \quad (3)$$

Energy equation for fluid domain:

$$\frac{\partial}{\partial x_j}(\rho_f U_j c_{p,f} T_f) = \frac{\partial}{\partial x_j} \left[\left(k_f + \frac{c_{p,f} \mu_t}{Pr_t} \right) \frac{\partial T_f}{\partial x_j} \right] \quad (4)$$

Energy equation for solid domain:

$$k_s \frac{\partial^2 T_s}{\partial x_j \partial x_j} = 0 \quad (5)$$

The turbulent viscosity, μ_t , in Eqs. (3) and (4) can be determined from the solution of the following transport equations for the turbulent kinetic energy, k , and the turbulent dissipation rate, ϵ , respectively,

Turbulent kinetic energy equation:

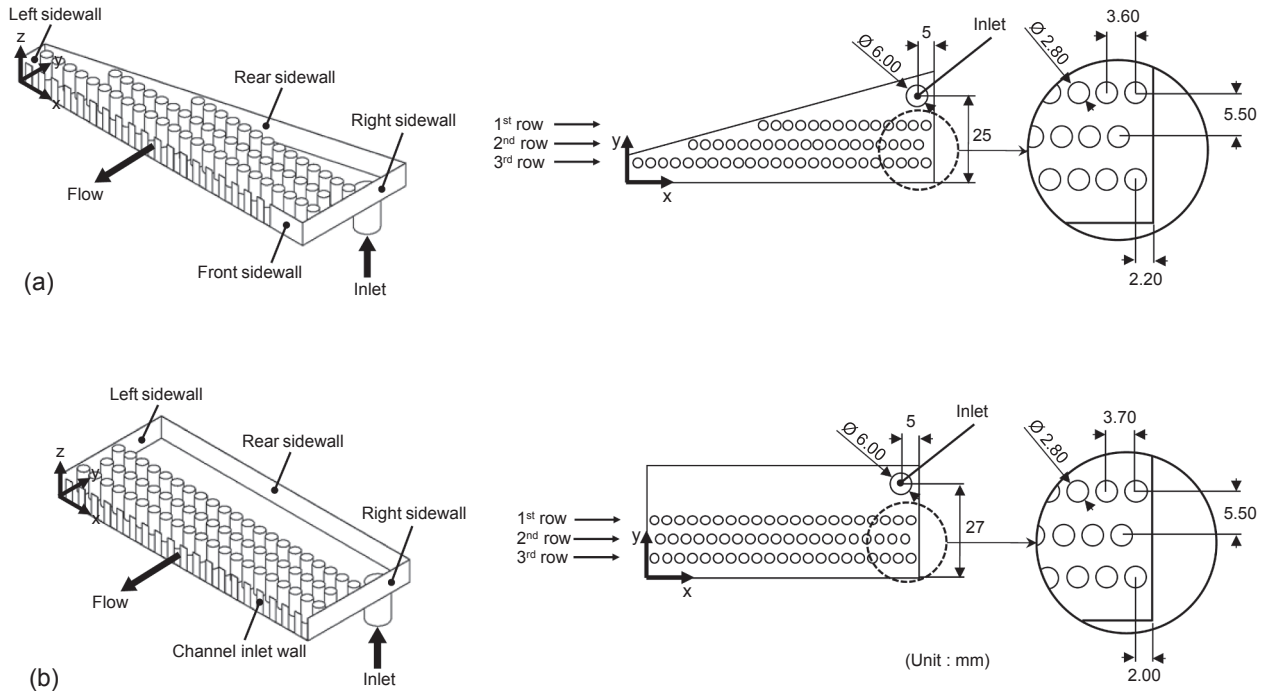


Fig. 2. Staggered pin-fin arrays within (a) trapezoidal and (b) rectangular inlet headers with pin-fin height of 6 mm.

$$\rho \frac{\partial}{\partial x_i} (k U_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k - Y_k \quad (6)$$

Turbulent dissipation rate equation:

$$\rho \frac{\partial}{\partial x_i} (\varepsilon U_i) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] + \rho C_1 S \varepsilon - \rho C_2 \frac{\varepsilon^2}{k + \sqrt{\nu \varepsilon}} \quad (7)$$

In Eqs. (6) and (7), the production of k , G_k , and the dissipation of k , Y_k are defined, respectively, as

$$G_k = \mu_t S^2 \quad (8)$$

$$Y_k = \rho \varepsilon \quad (9)$$

Empirical constants applied in Eqs. (6) and (7) are $\sigma_k = 1.0$, $\sigma_\varepsilon = 1.2$, $C_2 = 1.9$, and C_1 is given by

$$C_1 = \max[0.43, \eta/(\eta + 5)] \quad (10)$$

The flow in the inlet connection tube can be assumed to be fully developed turbulent, and the turbulence intensity, I , can be estimated using the following empirical correlation for pipe flows [35]

$$I(\%) = 0.16(Re_{in})^{-1/8} \times 100, \quad (11)$$

where the Reynolds number in the inlet connection tube, Re_{in} , is defined as

$$Re_{in} = \frac{4Q_{in}}{\pi \mu D_{in}} \quad (12)$$

The working fluid and solid material used in this study were water and aluminum, respectively. Solid properties were assumed to remain constant while temperature-dependent properties were taken into account for the fluid. Solid properties used during simulations included $\rho_s = 2719 \text{ kg/m}^3$, $c_{p,s} = 871 \text{ J/kg-K}$, and $k_s = 202.4 \text{ W/m-K}$, whereas fluid properties are evaluated using REFPROP 9.1 software from NIST [38]. The mass flow rate of water at 20°C was imposed at the inlet connection tube, and pressure was assumed to be uniform and equal to the atmospheric value at the outlet connection tube. The no-slip condition was applied at all solid boundaries, and a uniform heat flux q'' was considered out of the sink base (see Fig. 1(b)). All other surfaces

exposed to surroundings were considered adiabatic.

2.3. Grid independence test and numerical validation

A grid independence test was performed on the heat sink model with trapezoidal header (layout T-11) using three different grid systems comprising 2.80, 7.01, and 12.6 million cells. The maximum mean absolute error of mass flow rate in channels and maximum difference in total pressure drop, ΔP_{tot} , among the three grid systems were less than 1.90% and 1.46%, respectively, thereby indicating grid independence. The grid system comprising 7.01 million cells was used in all subsequent simulations. Fig. 3 depicts the grid structure of the selected grid system for heat sink equipped with trapezoidal header (layout T-11), wherein mini-channels were constructed using structured hexahedral grids while the header comprised unstructured grids.

Validation assessments were performed for both heat sinks with trapezoidal (layout T-1) and rectangular (layout R-1) headers by comparing the computed pressure drop in channels with predictions of previous pressure drop correlations. The pressure drop in each channel was predicted using the following relation,

$$\Delta P_{ch} = \frac{f_{app} \rho \bar{U}^2 L_{ch}}{2D_h} \quad (13)$$

where $\bar{U}$ refers to the average velocity in each channel obtained from numerical simulation. The apparent friction factor for the laminar developing flow region, f_{app} , could be determined using the following relation by Copeland [39],

$$f_{app} = \frac{\left\{ \left[3.2 \left(\frac{L_{ch}}{Re D_h} \right)^{-0.57} \right]^2 + (f_d Re)^2 \right\}^{1/2}}{Re} \quad (14)$$

where the friction factor for the laminar fully-developed flow in rectangular channel, f_d , is given by Shah and London [40]

$$f_d Re = 24(1 - 1.3553\beta + 1.9467\beta^2 - 1.7012\beta^3 + 0.9564\beta^4 - 0.2537\beta^5). \quad (15)$$

The channel Reynolds number is expressed as

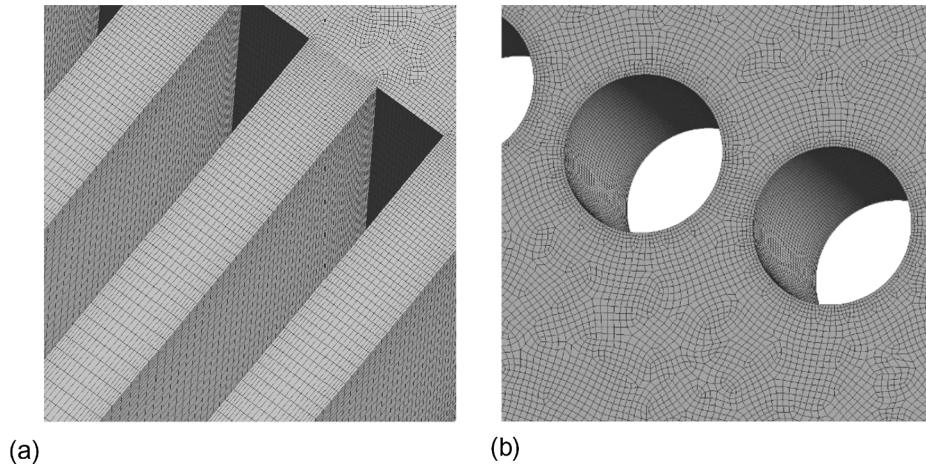


Fig. 3. Grid structure of fluid domain (a) near channel entrance and (b) around pin fins for T-11 layout (7.01×10^6 cells).

$$Re = \frac{\rho \bar{U} D_h}{\mu} \quad (16)$$

where D_h denotes the hydraulic diameter of mini-channel.

Fig. 4 shows comparison of computed and predicted pressure drops in each channel for both heat sinks with mass flow rate in the inlet connection tube of 68.3 g/s. Good agreement was observed between the three-dimensional numerical simulations and predictions of previous pressure drop correlations, with the average difference in channel pressure drop, ΔP_{ch} , of 3.6% and 4.0% for the trapezoid- and rectangle-shaped headers, respectively.

2.4. Design procedure

Numerical simulations were first performed on baseline layouts of the trapezoidal (T-1) and rectangular (R-1) headers to obtain a detailed understanding of the flow and pressure drop characteristics that cause flow maldistribution. Guided by the findings of baseline results, various pin-fin arrangements were subsequently attempted for each inlet header to minimize flow maldistribution within parallel channels. The general idea behind selection of pin-fin arrangements was to uniformly distribute the flow rate entering the mini-channels by suppressing undesirable flow features, such as strong circulating or transverse flows observed in the inlet header. For each layout, the pin-fin arrangement was modified by trial and error until a balanced flow distribution within parallel channels was realized. Flow maldistribution within parallel channels was quantitatively evaluated using the

maldistribution factor, MF , defined as

$$MF(\%) = \sqrt{\frac{1}{n-1} \sum_{k=1}^n \left(\frac{Q_k - Q_{avg}}{Q_{avg}} \right)^2} \times 100 \quad (17)$$

where the mass flow rate averaged over the entire mini-channels, Q_{avg} , is given by

$$Q_{avg} = \sum_{k=1}^n \frac{Q_k}{n}, \quad (18)$$

and Q_k denotes the mass flow rate in the k th channel with channel number ranging from 1 to 20, as depicted in Fig. 1(c). The maldistribution factor takes the form of the standard deviation of normalized mass flow rates in channels. Thus, a smaller value of the maldistribution factor indicates realization of better flow uniformity within channels.

3. Result and discussion

3.1. Effects of different pin-fin arrangements for trapezoidal inlet header

Ten different pin-fin arrangements for the trapezoid-shaped inlet header were designed to examine flow uniformity within parallel channels. Table 1 summarizes detailed design of the trapezoidal inlet header along with computed velocity contours at the mid-plane ($z = 3$ mm), maldistribution factor MF , and total pressure drop ΔP_{tot}

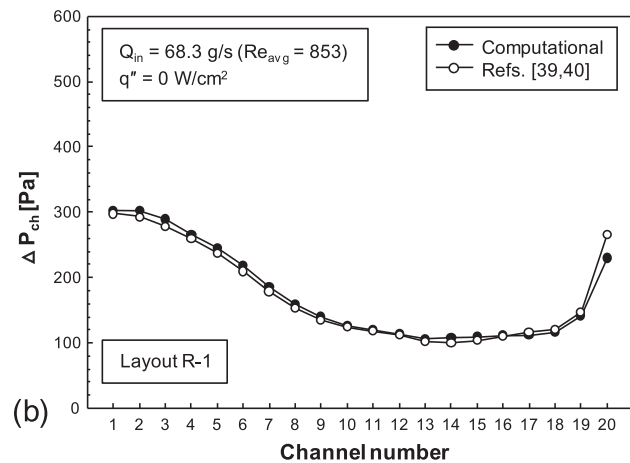
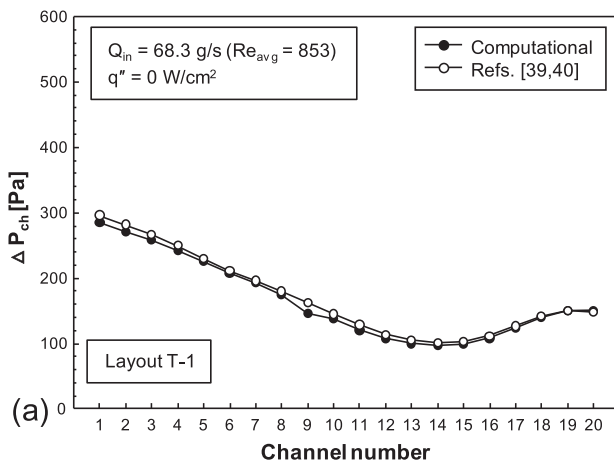
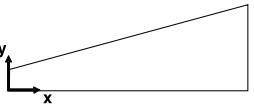
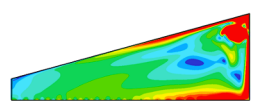
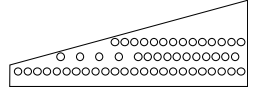
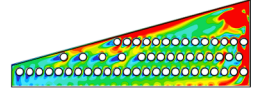
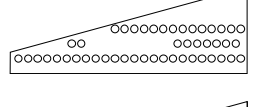
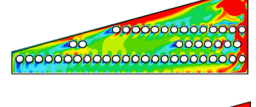
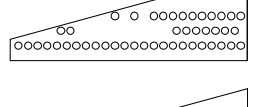
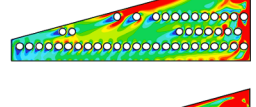
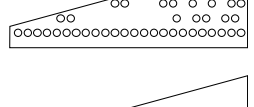
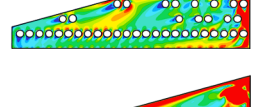
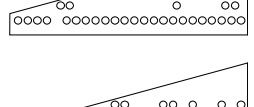
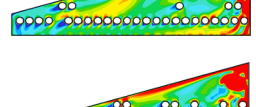
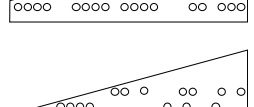
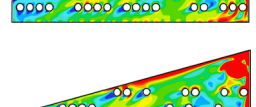
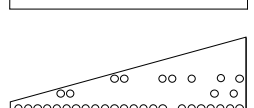
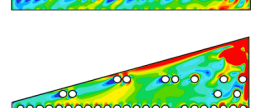
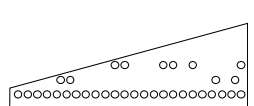
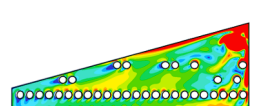
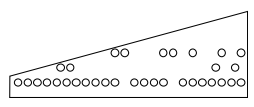
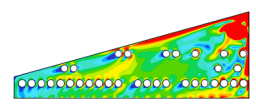
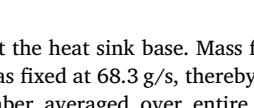
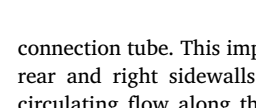


Fig. 4. Comparison of computed and predicted pressure drops across each channel of heat sink for (a) trapezoidal and (b) rectangular headers with baseline layout (without pin fin).

Table 1

Eleven different designs of trapezoidal inlet headers utilizing pin fins demonstrating computed velocity contour, maldistribution factor, and total pressure drop. Mass flow rate in inlet connection tube equaled 68.3 g/s ($Re_{avg} = 853$), and velocity contours were plotted at mid-plane ($z = 3$ mm).

Layout	Pin fin arrangement	Number of pin fin	Velocity magnitude (0 – 0.8 m/s)	MF (%)	ΔP_{tot} (Pa)
T-1 (w/o pin fins)		0		27.3	4880
T-2		53		10.5	5539
T-3		47		11.5	5440
T-4		45		9.1	5378
T-5		39		8.8	5312
T-6		36		9.1	5288
T-7		31		7.3	5353
T-8		30		6.6	5209
T-9		34		6.6	5229
T-10		34		6.4	5209
T-11 (optimized)		33		6.2	5232

without applying heat flux at the heat sink base. Mass flow rate in the inlet connection tube, Q_{in} , was fixed at 68.3 g/s, thereby corresponding to a channel Reynolds number averaged over entire mini-channels, Re_{avg} , of 853. To further understand flow distribution characteristics utilizing pin fins, velocity vectors in the inlet header (viewed from the top and cross-sections), as well as pressure contours corresponding to the inlet and outlet headers of the three layouts (T-1, T-2, and T-11) are presented in Fig. 5(a) and (b), respectively.

Velocity vectors of the baseline T-1 layout (without pin fins) at the mid-plane ($z = 3$ mm) are shown in Fig. 5(a). After entering the inlet header from the inlet connection tube, a high flow velocity region appears directly above the inlet connection tube due to the flow impingement effect attributed to vertical configuration of the inlet

connection tube. This impinging flow was subsequently confined to the rear and right sidewalls of the inlet header. The counter-clockwise circulating flow along the rear sidewall mostly enters the 1st to 3rd channels of the heat sink, whereas the clockwise circulating flow along the right sidewall impacts the front sidewall and gets subsequently diverted along the negative x-direction. This strong transverse flow along the negative x-direction near the entrance of mini-channels is indicated by arrows in Fig. 5(a). This offers an appreciable increase in the vena-contracta size on one side of each channel (i.e., the right sidewall of channel), as indicated by the dashed circle in Fig. 6, which in turn, contributes towards preventing the fluid from entering the mini-channels due to the reduced cross-sectional flow area. Also depicted in the figure is that the transverse flow velocity reduced as the flow proceeds

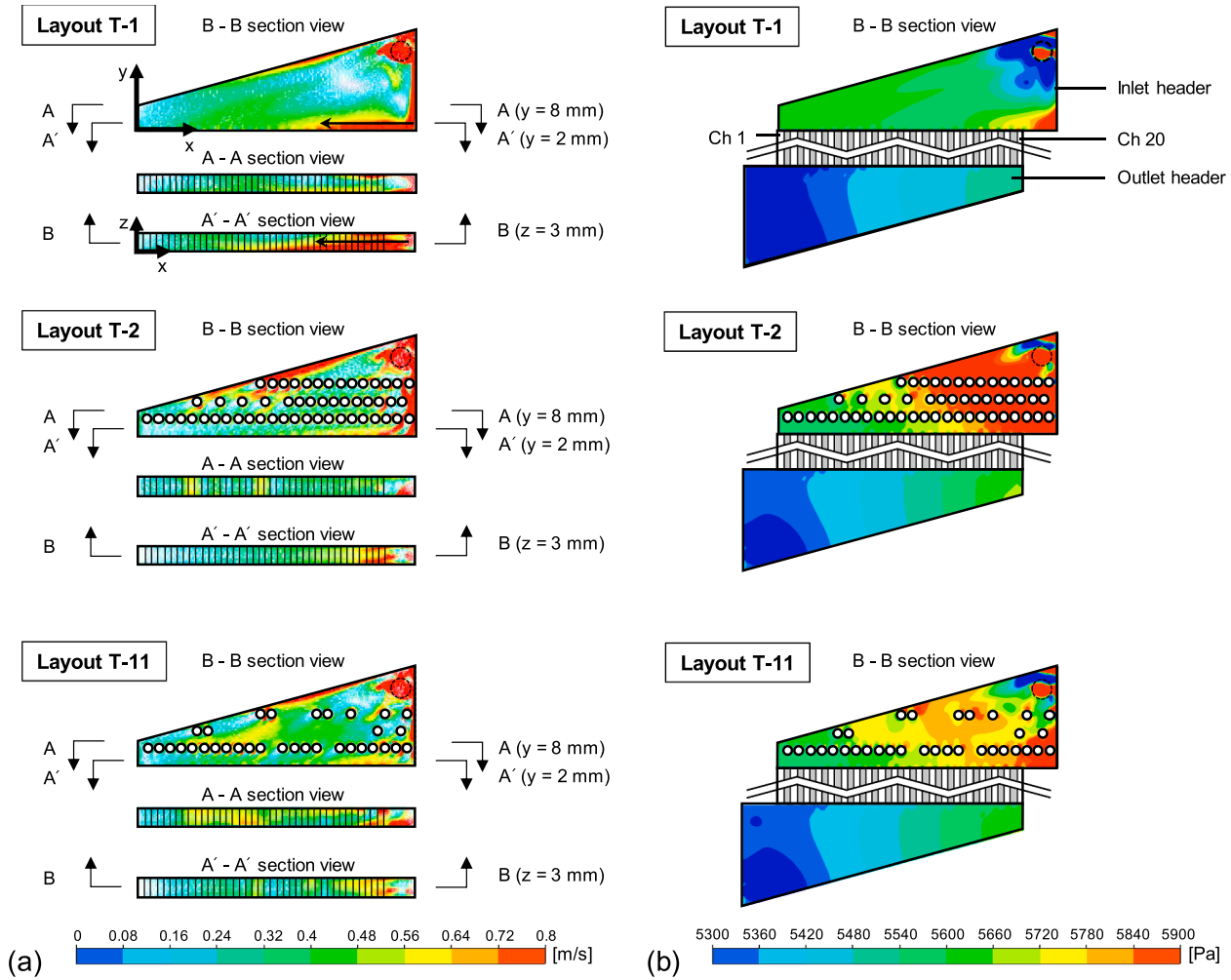


Fig. 5. (a) Velocity vectors for trapezoidal inlet header and (b) pressure contours for trapezoidal inlet and outlet headers with different layouts at $Q_{in} = 68.3$ g/s and $q'' = 0$ W/cm².

along the negative x-direction, thereby resulting in reduction of the vena-contracta size at the entrance of each channel in accordance with decreasing channel number. As a result, the flow rate in each channel generally increases as the channel number decreases. It should be noted, however, that despite the large vena-contracta size at the entrance of the 19th and 20th channels, a relatively large amount of fluid was distributed to the channels due to the flow impingement effect ascribed to the vertical configuration of the inlet connection tube. Previous studies [27,28] examining the flow impingement effect on the flow distribution within channels revealed that the impinging flow spreads out in every direction of the inlet header after impacting the base (xy-plane), and that each flow streamline directly follows the shortest flow path to the channel entrance. This explains existence of a relatively high flow rate in the 19th and 20th channels, which are located in close proximity to the inlet connection tube. For the same reason, the flow rate gradually decreases with decreasing channel number until it attains its lowest value in the 14th channel.

Overall, above-mentioned flow characteristics of the inlet header equipped with a trapezoidal heat sink are responsible for flow maldistribution in parallel channels corresponding to the T-1 layout. It is, therefore, necessary to place pin fins to facilitate inhibition of clockwise and counterclockwise circulating flows along sidewalls as well as the transverse flow adjacent to channel entrances. As shown in Figs. 5 and 6, with the aid of pin fins inserted in the inlet header, transverse flow appears to weaken in the T-2 layout and significantly subside in the T-11 layout. Consequently, the vena-contracta size in the 20th channel

was reduced in the T-11 layout, thus showing better flow uniformity within parallel channels compared to the baseline T-1 layout.

To better describe flow distribution characteristics of different trapezoidal-header designs, pressure contours at the mid-plane ($z = 3$ mm) of inlet and outlet headers are depicted in Fig. 5(b). Once flow enters into the inlet header from the inlet connection tube, a high-pressure region forms directly above the inlet connection tube, whereas a low-pressure region appears around the flow impingement area. The static pressure then gradually rises as fluid moves along the negative x-direction, whereas in the outlet header, pressure decreases along the negative x-direction as fluid moves towards the outlet connection tube. Therefore, the pressure difference between entry and exit from the 1st channel far exceeds that corresponding to the 20th channel, thereby resulting in higher and lower flow rates within the 1st and 20th channels, respectively. However, as illustrated in the pressure contour corresponding to the T-2 layout, static pressure in the right half of the inlet header is quite high due to the densely placed pin fins, thereby resulting in higher pressure drop in channels on the right of the heat sink, and hence, higher flow rate near the 20th channel. Comparison of pressure contours corresponding to the T-2 and T-11 layouts demonstrates smaller variation in pressure drop across parallel channels in the T-11 layout compared to the T-2 layout, which in turn leads to better flow uniformity in the T-11 layout.

Fig. 7 compares normalized mass flow rates of each channel for three different designs of the trapezoidal inlet header. These results indicate that the observed severely pronounced flow maldistribution in

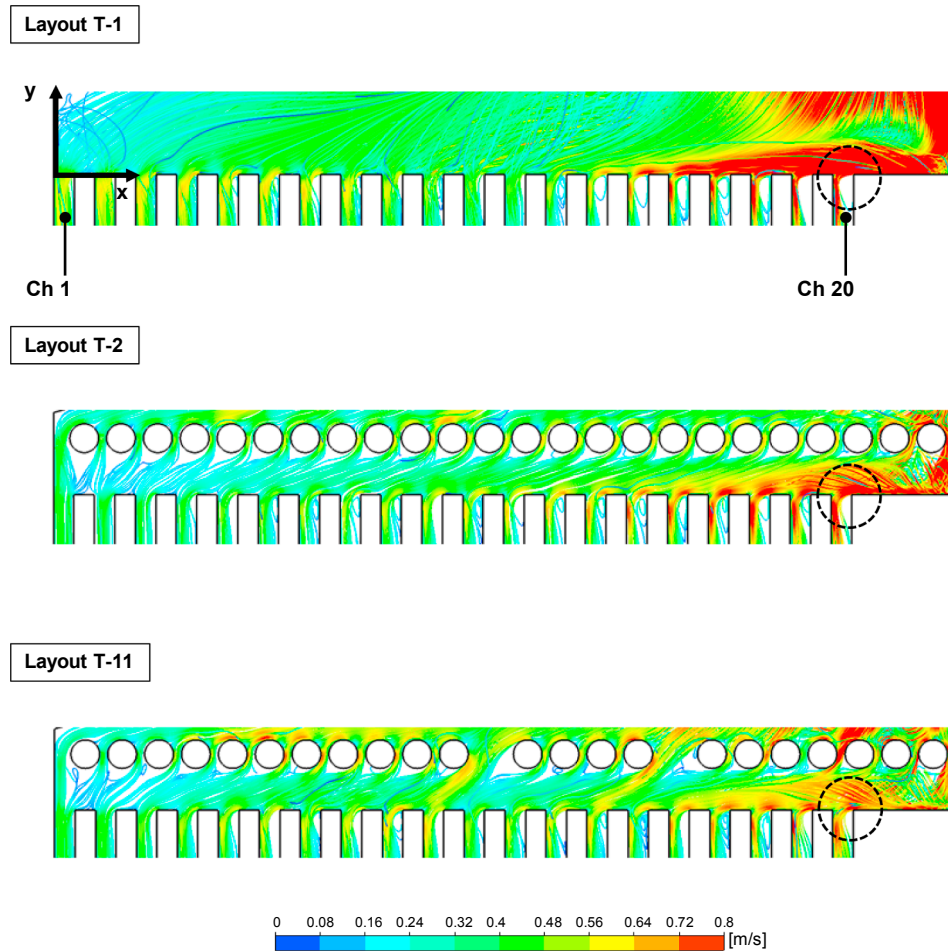


Fig. 6. Streamlines near channel entrance for trapezoidal inlet header with different layouts at $Q_{in} = 68.3$ g/s and $q'' = 0$ W/cm².

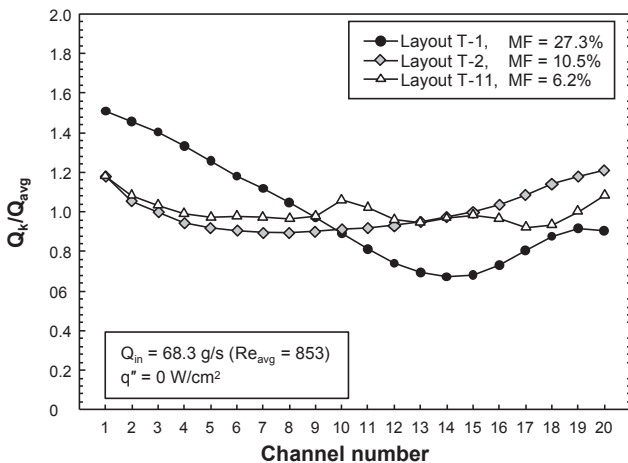


Fig. 7. Normalized mass flow rate of each channel for different layouts of trapezoidal inlet header.

the baseline T-1 layout considerably improves in the T-11 (“best”) layout by means of a suitable arrangement of pin fins in the inlet header, as evidenced by MF values of 27.3% and 6.2% corresponding to the T-1 and T-11 layouts, respectively.

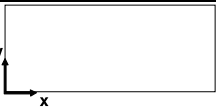
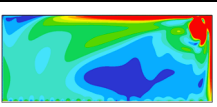
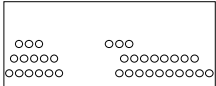
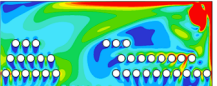
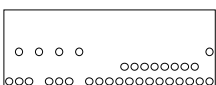
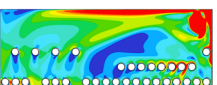
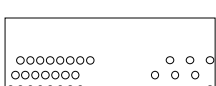
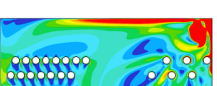
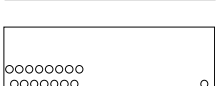
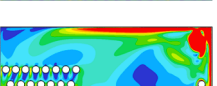
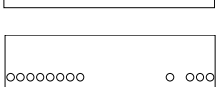
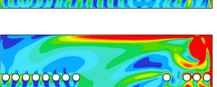
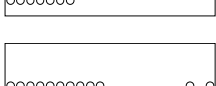
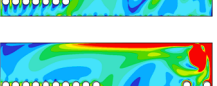
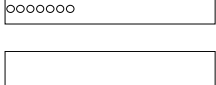
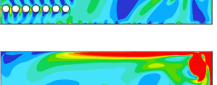
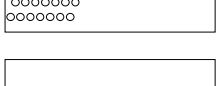
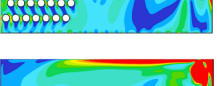
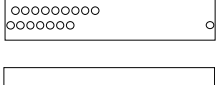
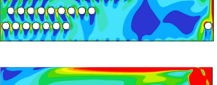
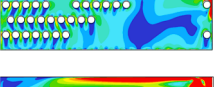
3.2. Effects of different Pin-fin arrangements for rectangular inlet header

Similar to the trapezoidal inlet header described in the previous

section, ten different pin-fin arrangements for the rectangular inlet-header configuration, as described in Table 2, were designed to examine flow uniformity within parallel channels. Velocity vectors for the baseline R-1 layout (without pin fins), plotted at the mid-plane ($z = 3$ mm), are depicted in Fig. 8(a). Similar to the trapezoidal inlet header, the flow velocity, after entry into the inlet header from the inlet connection tube, is high along the rear and right sidewalls due to flow confinement by the sidewalls. The flow along the rear sidewall then separates from the wall surface due to the corner vortex generated in the upper-left corner of the inlet header. Subsequently, the flow, in most likelihood, enters channels 1–4 of the heat sink, as described by velocity vectors in Fig. 8(a). On the other hand, flow along the right sidewall directly enters the 20th channel. It should be noted that the strong transverse flow along the negative x-direction near channel entrances, which was observed in the case of the trapezoidal header in the baseline T-1 layout, no longer occurs due to geometrical differences that exist between the two header types in terms of absence of the front sidewall in the rectangular inlet header. Another distinct flow feature of the R-1 layout is the creation of large recirculation zones in the middle of the inlet header comprising two counter-rotating vortices, as illustrated by arrows in Fig. 8(a). Previous studies [21–24,32] examining the effect of recirculation zones in heat sink headers have reported that existence of a recirculation zone contributes to poor flow uniformity within channels. This can be confirmed by the existence of a relatively large vena-contracta size in channel 11, as indicated by the dashed circle in Fig. 9, which is directly affected by flow recirculation near the channel entrance. Existence of a pin-fin arrangement can, therefore, be considered to reduce flow rates in channels 1–4 and 20 whilst correspondingly increasing flow rates in the middle channels by removing

Table 2

Eleven different designs of rectangular inlet headers utilizing pin fins demonstrating computed velocity contour, maldistribution factor, and total pressure drop. Mass flow rate in inlet connection tube equaled 68.3 g/s ($Re_{avg} = 853$), and velocity contours were plotted at mid-plane ($z = 3$ mm).

Layout	Pin fin arrangement	Number of pin fin	Velocity magnitude (0–0.8 m/s)	MF (%)	ΔP_{tot} (Pa)
R-1 (w/o pin fins)		0		30.9	4867
R-2		35		19.7	4941
R-3		32		25.2	4916
R-4		30		14.8	4865
R-5		37		11.3	5008
R-6		29		14.2	4891
R-7		27		14.1	4897
R-8		28		11.0	5030
R-9		28		8.4	4935
R-10		28		9.0	4911
R-11 (optimized)		33		7.1	4896

recirculation zones. As shown in the R-2 layout, 35 pin fins were inserted on the left and right sides of the inlet header. As a result, flow entering channels 1–4 appears to have substantially weakened, whereas a large amount of fluid was observed to flow through passage between adjacent pin fins wherein flow resistance is small. This contributes to a significant increase in flow rates through channels 6–8. To facilitate even distribution of flow rates within channels, various pin-fin arrangements in the inlet header, as described in Table 2, were attempted to arrive at the best layout, R-11. As illustrated in Fig. 8(a), more pin fins were placed in the first row in the middle of the inlet header to reduce higher flow rates at that location, as observed in the R-2 layout. As a result, the vena-contracta size was substantially reduced in the R-11 layout, as depicted in Fig. 9. In addition, the two pin fins inserted

into the right end of the first row in the inlet header contributed to effectively reduce the flow rate entering the 20th channel.

Flow distribution characteristics of the heat sink for different designs of the rectangular inlet header were also investigated by plotting pressure contours of the inlet and outlet headers at the mid-plane ($z = 3$ mm), as depicted in Fig. 8(b). For the baseline layout (R-1), the pressure difference between entry and exit was observed to be the largest in the 1st channel, and the same demonstrated a decrease with increasing channel number up to the 14th channel. In the R-2 layout, however, the largest pressure difference was observed in the 7th channel. For both layouts (R-1 and R-2), the channel pressure drop was observed to be fairly high in the 20th channel. The observed variation in channel pressure drop across parallel channels was least in the case

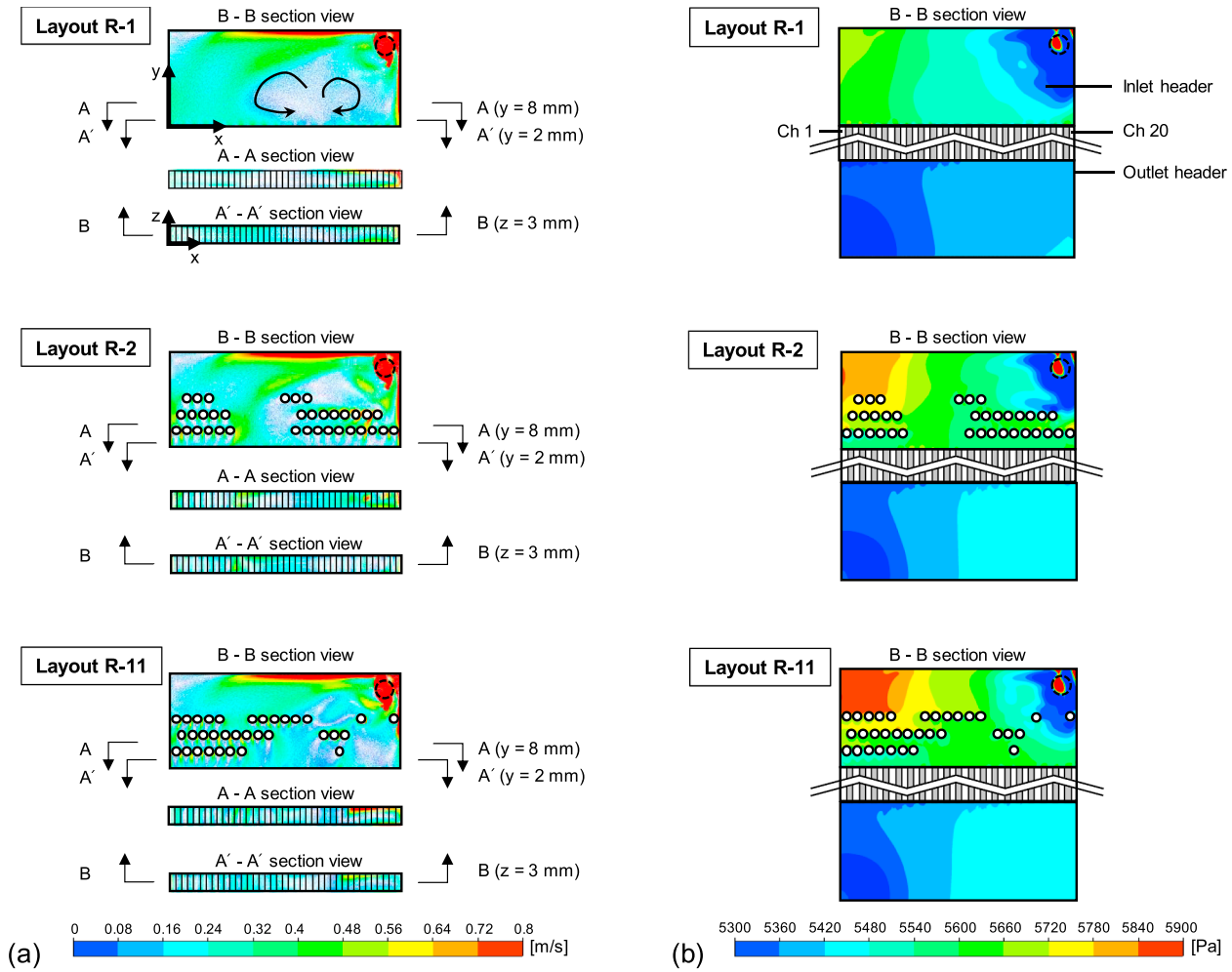


Fig. 8. (a) Velocity vectors for rectangular inlet header and (b) pressure contours for rectangular inlet and outlet headers with different layouts at $Q_{in} = 68.3$ g/s and $q'' = 0$ W/cm².

of the best layout (R-11), thereby resulting in better flow uniformity within channels.

Fig. 10 compares normalized mass flow rates of each channel for three different designs of the rectangular inlet header, thereby confirming that a higher channel pressure drop leads to higher flow rate through the channel. This result shows that the R-11 layout provides the most uniform flow distribution, as evidenced by the corresponding smallest MF value of 7.1%.

Tables 1 and 2 list values of the total pressure drop between the inlet and outlet connection tubes, ΔP_{tot} , for different designs of the trapezoidal and rectangular inlet headers, respectively. It should be noted that increase in ΔP_{tot} due to the use of pin fins is small; ΔP_{tot} difference between the T-1 (without pin fins) and T-11 (with pin fins) layouts is 7.2%, whereas that between the R-1 and R-11 layouts is only 0.6%. This indicates that the use of pin-fin arrays in the inlet header is a promising method for improving flow uniformity within parallel channels, particularly with respect to avoiding a significant increase in total pressure drop.

3.3. Effects of inlet mass flow rate

To examine the effect of different mass flow rates in the inlet tube on flow distribution, normalized mass flow rates of each channel in the best trapezoidal and rectangular header layouts were obtained from numerical simulations, as depicted in Fig. 11(a) and (b), respectively. Three different mass flow rates in the inlet tube, Q_{in} , of 34.1, 68.3, and

102.4 g/s, corresponding to Re_{avg} values of 427, 853, and 1280, respectively, were compared. The result obtained indicated that the flow distribution within channels corresponding to the T-11 trapezoidal layout remained nearly unaffected by variation in Q_{in} , as evidenced by the largest MF value of 7.9% among all cases. As regards the case of the R-11 rectangular layout, flow rates in the 19th and 20th channels were observed to slightly decrease with increasing Q_{in} . This could be attributed to the large contraction pressure loss at the inlet of the 19th and 20th channels owing to existence of a relatively high flow velocity on the right of the inlet header, wherein only a few pin fins were placed, compared to those existing on the left of the inlet header.

Fig. 12(a) compares MF values obtained for the baseline and best layouts for different Re_{avg} values. In both cases, the MF value demonstrated increases with increasing Re_{avg} . It is important to note that flow uniformity within mini-channel heat sinks greatly improves upon utilization of pin fins in both the trapezoidal and rectangular inlet-header configurations regardless of the value of Re_{avg} , thereby resulting in far lesser values of MF for the best layouts (T-11 and R-11) compared to their respective baselines (T-1 and R-1). Fig. 12(b) shows ΔP_{tot} values corresponding to the baseline and best layouts for different Re_{avg} values. Once again, in both cases, the value of ΔP_{tot} demonstrated increase with increase in Re_{avg} . However, the use of pin fins in the inlet header demonstrated minimal influence on the total pressure drop; the largest observed value of the ΔP_{tot} difference between the baseline and best layouts was only 8.1% for the trapezoidal header with $Re_{avg} = 1280$. It should be noted that for the rectangular-header case, the observed

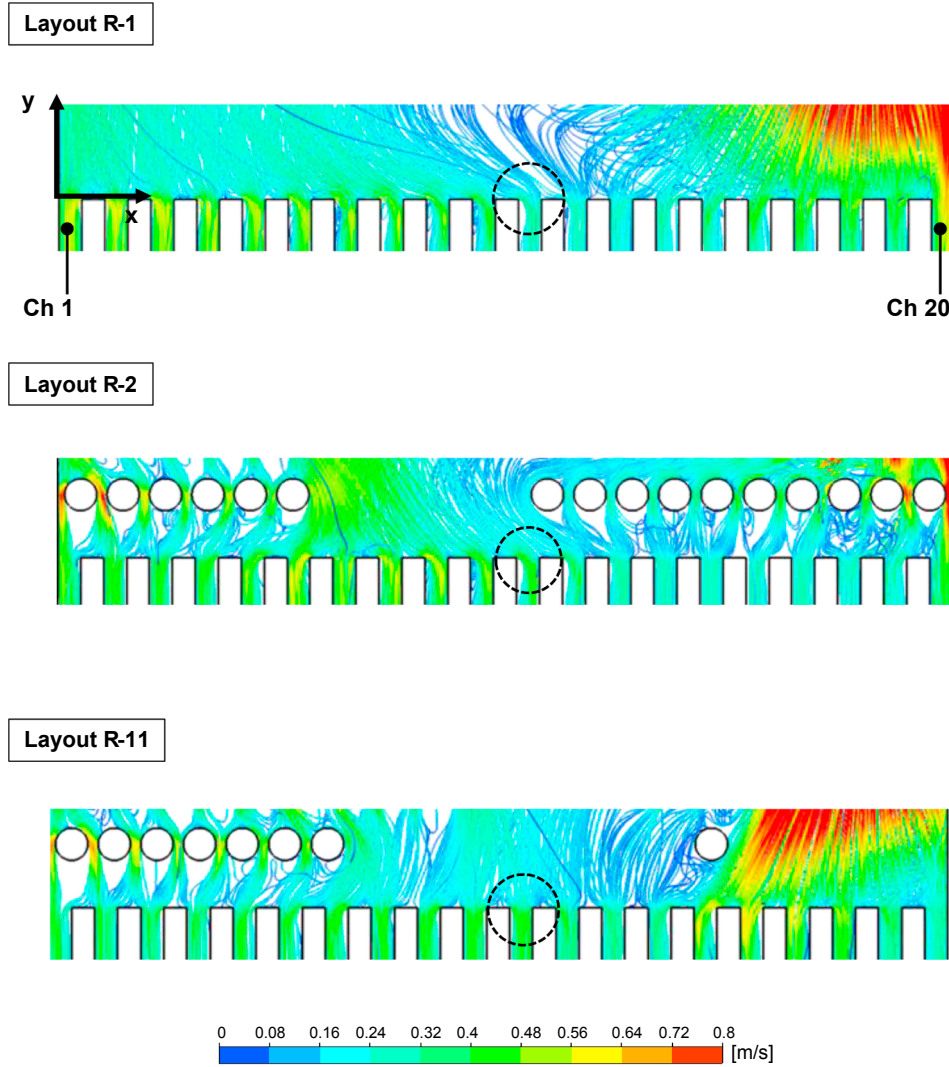


Fig. 9. Streamlines near channel entrance for rectangular inlet header with different layouts at $Q_{in} = 68.3$ g/s and $q'' = 0$ W/cm².

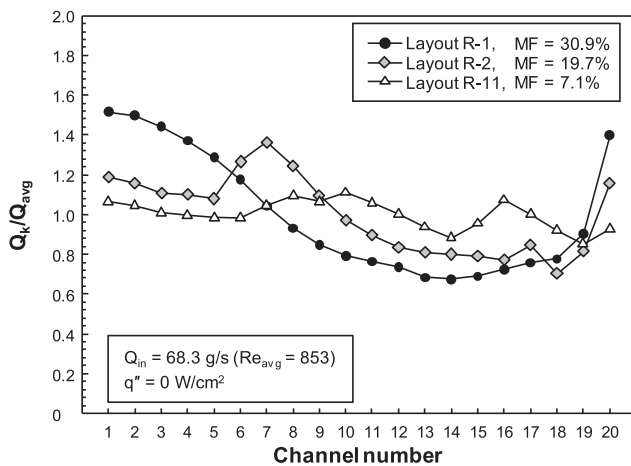


Fig. 10. Normalized mass flow rate of each channel for different layouts of rectangular inlet header.

increase in ΔP_{tot} owing to the use of pin fins was rather small with the largest ΔP_{tot} difference between the baseline and best layouts being of the order of 1.2% with $Re_{avg} = 427$.

3.4. Effects of heat sink design on thermal performance

The influence of flow uniformity in parallel mini-channels on heat transfer performance of heat sinks was also investigated in this study. Thermal characteristics of heat sinks for the different layouts were analyzed using two base heat flux (q'') values of 7 and 15 W/cm². Representative temperature contours corresponding to the baseline and best layouts have been compared in Fig. 13, reference to which demonstrates that for all layouts, the heat sink wall temperature is rather low at the channel entrance due to a high value of the heat transfer coefficient associated with the hydrodynamic and thermally developing flow, and generally increases along the flow direction. For both baseline layouts (T-1 and R-1), a high-temperature region was observed to appear downstream of channels 12–17, wherein a relatively small amount of fluid was distributed. In the case of the best layouts (T-11 and R-11), however, due to enhanced flow uniformity within channels, the temperature distribution over the heat sink base was observed to become uniform compared to respective baseline cases. This temperature-distribution uniformity and thermal performance of the heat sink could be quantified by introducing the maximum temperature difference of the heat sink base (ΔT_b) and overall heat sink thermal resistance (R_{th}), both of which could, respectively, be expressed as

$$\Delta T_b (^{\circ}\text{C}) = T_{b,max} - T_{b,min} \quad (19)$$

and

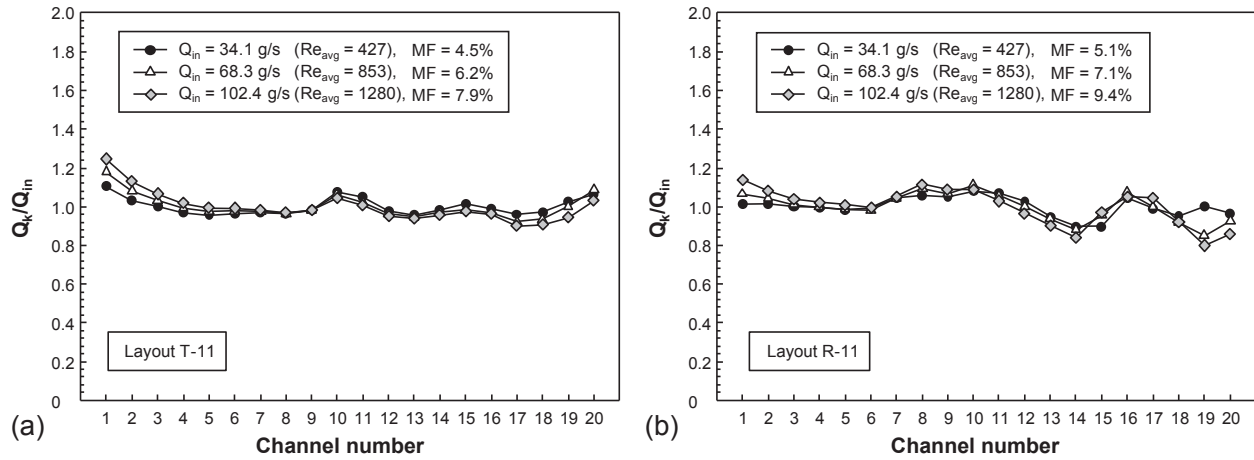


Fig. 11. Effect of different mass flow rates in inlet tube on flow distribution for (a) trapezoidal and (b) rectangular inlet headers with best layout at $q'' = 0 \text{ W/cm}^2$.

$$R_{th} (\text{°C/W}) = \frac{T_{b,max} - T_{in}}{q'' A_b} \quad (20)$$

where $T_{b,max}$ and $T_{b,min}$ in Eq. (19) denote the observed maximum and minimum temperatures of the heat sink base, respectively, and T_{in} in Eq. (20) denotes fluid temperature at the inlet connection tube.

Fig. 14 compares the maximum temperature differences of the heat sink base ΔT_b between the baseline and best layouts for different heat fluxes and mass flow rates. Fig. 14(a) and (b) confirms that regardless of the value of Q_{in} and q'' , temperature-distribution uniformity is greatly improved by enhancing flow uniformity within channels. This could be attributed to utilization of a suitable pin-fin arrangement in trapezoidal and rectangular inlet headers. The observed average difference in ΔT_b values between T-1 and T-11 layouts equaled 15.7%, while that between R-1 and R-11 equaled 14.9%.

Fig. 15 compares the observed overall thermal resistance R_{th} between the baseline and best layouts for different heat fluxes and mass flow rates. Fig. 15(a) and (b) shows that for both the baseline and best layouts of the trapezoidal and rectangular header, R_{th} decreases with increasing Q_{in} while q'' demonstrates only a minor influence on R_{th} . It should be noted that a noticeable improvement in heat-sink thermal performance is realized for best layouts of both the trapezoidal and rectangular headers compared to respective baseline layouts. The observed average difference in values of R_{th} between the T-1 and T-11 layouts equaled 11.8%, while that between layouts R-1 and R-11 equaled 13.0%.

3.5. Future work

The proposed study focused on the development of V-type heat sinks equipped with two different types of headers, trapezoidal and rectangular. To eliminate undesirable flow features observed during baseline simulations that result in flow maldistribution within parallel channels, various pin-fin arrangements were intuitively attempted in the inlet header configuration for each layout. However, the design procedure for determining the most suitable arrangement of pin fins described in this study is one involving trial and error, and the same is applicable only to present V-type heat sinks. Therefore, further research endeavors need to be undertaken to develop a systematic methodology for optimization of the shape, size, and arrangement of pin fins with due consideration of various geometrical and operating parameters concerning heat sinks.

Overall, although limited in scope and parameter range, the present study provides valuable insight into complicated flow distribution characteristics of heat sinks. The study also demonstrates the effectiveness of installing pin fins in the inlet header of mini-channel heat sinks in terms of enhanced flow uniformity within parallel channels. However, validation of numerical simulations can greatly benefit from detailed flow-velocity measurements performed within each channel in future endeavors along with flow visualization within the inlet header. As discussed in [41,42], such measurements are possible with the aid of microparticle image velocimetry (μ -PIV). For successful application of heat sinks with pin-finned inlet headers in high-heat-flux thermal

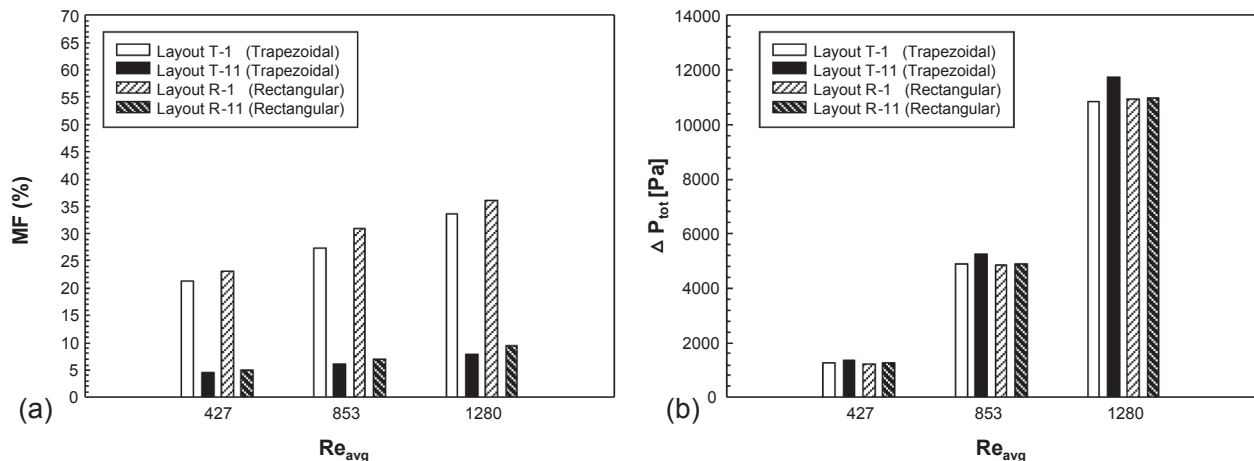


Fig. 12. Comparison of (a) maldistribution factor and (b) total pressure drop between baseline and best layouts for different average channel Reynolds numbers at $q'' = 0 \text{ W/cm}^2$.

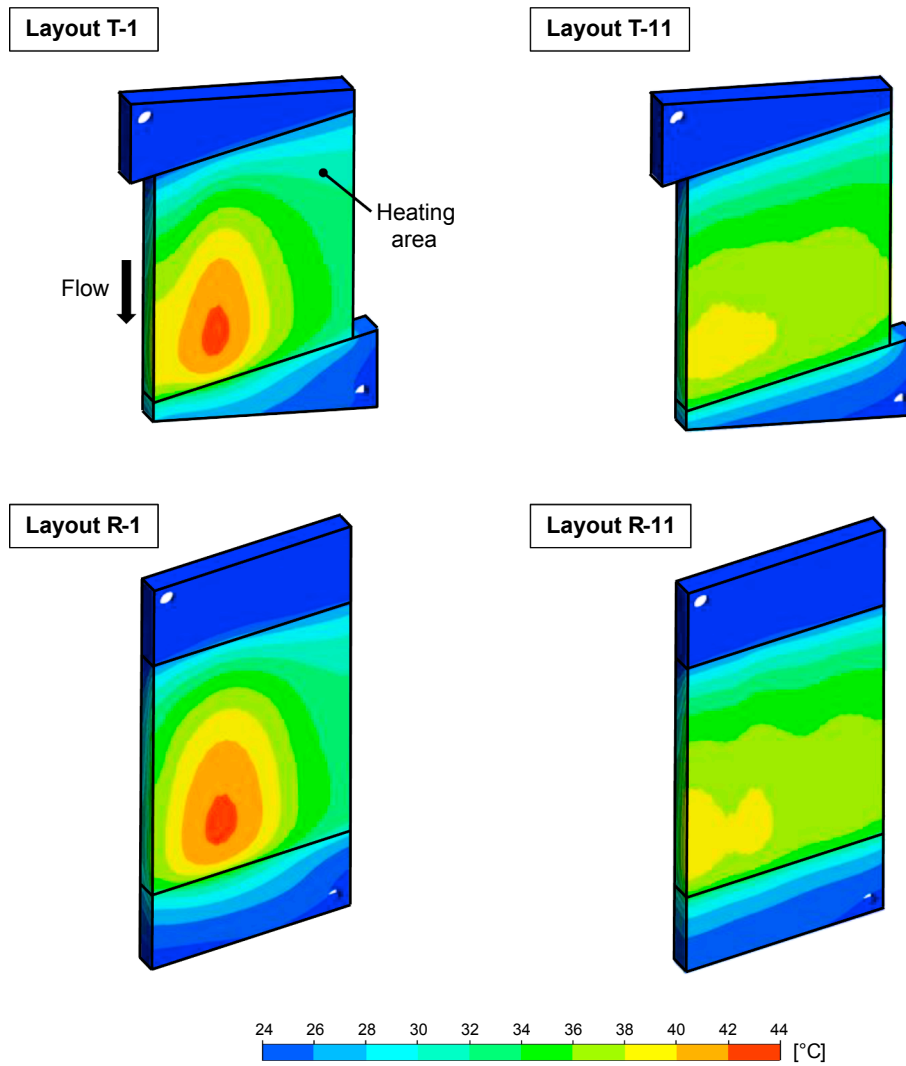


Fig. 13. Temperature contours for mini-channel heat sinks with different layouts at $Q_{in} = 102.4$ g/s and $q'' = 15$ W/cm².

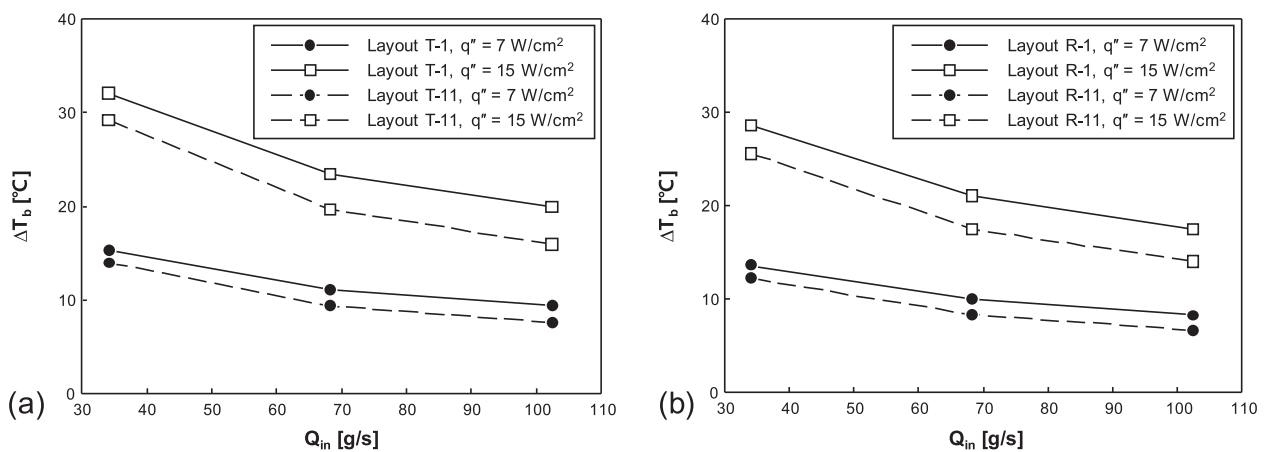


Fig. 14. Comparison of temperature distribution uniformity between baseline and best layouts for (a) trapezoidal and (b) rectangular headers.

management, efforts must also be made to address the relationship between flow distribution, flow instabilities, flow-induced vibrations, and non-uniform heat-flux distributions.

4. Conclusions

The present study concerns the development of a new strategy utilizing staggered pin-fin arrays in the inlet header of mini-channel heat sinks aimed at achieving uniform flow distribution within parallel channels without increasing total pressure drop. Two different header

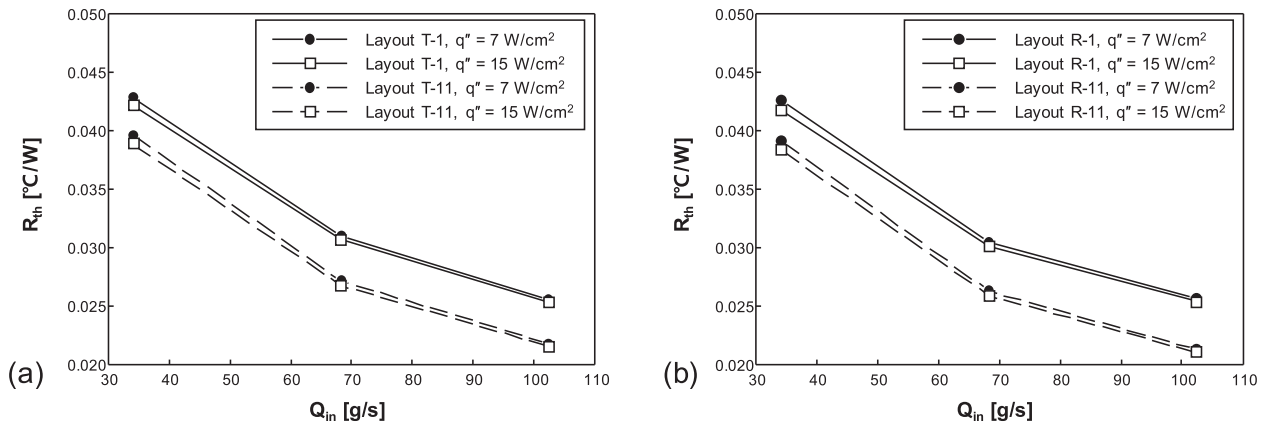


Fig. 15. Comparison of overall heat-sink thermal resistance between baseline and best layouts for (a) trapezoidal and (b) rectangular headers.

shapes, trapezoidal and rectangular, were considered, and ten different pin-fin arrangements for each header shape were designed to numerically investigate flow uniformity within channels. Key findings from the study are as follows:

- (1) Flow maldistribution within mini-channel heat sinks greatly improves via use of a suitable arrangement of pin fins in the inlet header. As a result, best layouts corresponding to the trapezoidal (T-11) and rectangular (R-11) headers were observed to provide the most uniform flow distribution among the eleven different designs considered for each header type, as evidenced by the smallest *MF* values of 6.2% and 7.1%, respectively.
- (2) Comparison of *MF* values corresponding to the baseline and best layouts for different Re_{avg} values demonstrated that *MF* values increase with increasing Re_{avg} , and that the flow uniformity within mini-channel heat sinks is greatly improved by utilizing pin fins regardless of the value of Re_{avg} . This results in realization of smaller values of *MF* for best layouts (T-11 and R-11) compared to respective baseline cases (T-1 and R-1).
- (3) Values of ΔP_{tot} corresponding to the baseline and best layouts for different Re_{avg} values were compared. For both baseline and best layouts, values of ΔP_{tot} were observed to increase with increase in Re_{avg} . The use of pin fins in the inlet header demonstrated minimal influence on the total pressure drop. The largest difference in the ΔP_{tot} value between the baseline and best layouts equaled 8.1% when employing a trapezoidal header with $Re_{avg} = 1280$ and only 1.2% for the rectangular header with $Re_{avg} = 427$.
- (4) Comparison of ΔT_b and R_{th} corresponding to the baseline and best layouts for different q'' and Q_{in} revealed that regardless of q'' and Q_{in} values, a noticeable improvement in temperature-distribution uniformity and thermal performance was achieved upon utilization of suitable pin-fin arrangements in the best layouts. The average difference in ΔT_b and R_{th} values between the baseline and best layouts equaled 15.7% and 11.8%, respectively, for the trapezoidal header, and 14.9% and 13.0%, respectively, for the rectangular header.

Acknowledgement

This study was supported by Agency for Defense Development (ADD).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2019.02.069>.

References

- [1] I. Mudawar, Assessment of high-heat-flux thermal management schemes, *IEEE Trans. Compon. Packag. Technol.* 24 (2001) 122–141.
- [2] R. Chein, G. Huang, Analysis of microchannel heat sink performance using nano-fluids, *Appl. Therm. Eng.* 25 (2005) 3104–3114.
- [3] R. Revellin, J.R. Thome, A new type of diabatic flow pattern map for boiling heat transfer in microchannels, *J. Micromech. Microeng.* 17 (2007) 788–796.
- [4] L. Wojtan, T. Ursenbacher, J.R. Thome, Investigation of flow boiling in horizontal tubes: Part 1 – a new diabatic two-phase flow pattern map, *Int. J. Heat Mass Transfer* 48 (2005) 2955–2969.
- [5] A. Kundu, R. Kumar, A. Gupta, Heat transfer characteristics and flow pattern during two-phase flow boiling of R134a and R407C in a horizontal smooth tube, *Exp. Therm. Fluid Sci.* 57 (2014) 344–352.
- [6] H. Zhang, I. Mudawar, M.M. Hasan, CHF model for subcooled flow boiling in Earth gravity and microgravity, *Int. J. Heat Mass Transfer* 50 (2007) 4039–4051.
- [7] H. Lee, I. Mudawar, M.M. Hasan, Experimental and theoretical investigation of annular flow condensation in microgravity, *Int. J. Heat Mass Transfer* 61 (2013) 293–309.
- [8] K. Balasubramanian, M. Jagirdar, P.S. Lee, C.J. Teo, S.K. Chou, Experimental investigation of flow boiling heat transfer and instabilities in straight microchannels, *Int. J. Heat Mass Transfer* 66 (2013) 655–671.
- [9] S. Lee, V.S. Devahdhanush, I. Mudawar, Frequency analysis of pressure oscillations in large length-to-diameter two-phase micro-channel heat sinks, *Int. J. Heat Mass Transfer* 116 (2018) 273–291.
- [10] S.M. Kim, I. Mudawar, Theoretical model for local heat transfer coefficient for annular flow boiling in circular mini/micro-channels, *Int. J. Heat Mass Transfer* 73 (2014) 731–742.
- [11] S.M. Kim, I. Mudawar, Theoretical model for annular flow condensation in rectangular micro-channels, *Int. J. Heat Mass Transfer* 55 (2012) 958–970.
- [12] S.M. Kim, I. Mudawar, Universal approach to predicting two-phase frictional pressure drop for mini/micro-channel saturated flow boiling, *Int. J. Heat Mass Transfer* 58 (2013) 718–734.
- [13] S.M. Kim, I. Mudawar, Universal approach to predicting two-phase frictional pressure drop for adiabatic and condensing mini/micro-channel flows, *Int. J. Heat Mass Transfer* 55 (2012) 3246–3261.
- [14] S.M. Kim, I. Mudawar, Universal approach to predicting saturated flow boiling heat transfer in mini/micro-channels – Part II. Two-phase heat transfer coefficient, *Int. J. Heat Mass Transfer* 64 (2013) 1239–1256.
- [15] S.M. Kim, I. Mudawar, Universal approach to predicting heat transfer coefficient for condensing mini/micro-channel flow, *Int. J. Heat Mass Transfer* 56 (2013) 238–250.
- [16] S.M. Kim, I. Mudawar, Universal approach to predicting saturated flow boiling heat transfer in mini/micro-channels – Part I. Dryout incipience quality, *Int. J. Heat Mass Transfer* 64 (2013) 1226–1238.
- [17] S.M. Kim, I. Mudawar, Thermal design and operational limits of two-phase micro-channel heat sinks, *Int. J. Heat Mass Transfer* 106 (2017) 861–876.
- [18] S.M. Kim, I. Mudawar, Review of two-phase critical flow models and investigation of the relationship between choking, premature CHF, and CHF in micro-channel heat sinks, *Int. J. Heat Mass Transfer* 87 (2015) 497–511.
- [19] J.B. Kitto Jr, J.M. Robertson, Effects of maldistribution of flow on heat transfer equipment performance, *Heat Transfer Eng.* 10 (1989) 18–25.
- [20] S.M. Kim, I. Mudawar, Review of databases and predictive methods for heat transfer in condensing and boiling mini/micro-channel flows, *Int. J. Heat Mass Transfer* 77 (2014) 627–652.
- [21] R. Chein, J. Chen, Numerical study of the inlet/outlet arrangement effect on microchannel heat sink performance, *Int. J. Therm. Sci.* 48 (2009) 1627–1638.
- [22] M.C. Lu, C.C. Wang, Effect of the inlet location on the performance of parallel-channel cold-plate, *IEEE Trans. Compon. Packag. Technol.* 29 (2006) 30–38.
- [23] R.M. Kumaran, G. Kumaraguruparan, T. Sornakumar, Experimental and numerical studies of header design and inlet/outlet configurations on flow mal-distribution in parallel micro-channels, *Appl. Therm. Eng.* 58 (2013) 205–216.

- [24] J.C. Tong, E.M. Sparrow, J.P. Abraham, Geometric strategies for attainment of identical outflows through all of the exit ports of a distribution manifold in a manifold system, *Appl. Therm. Eng.* 29 (2009) 3552–3560.
- [25] V. Manoj Siva, A. Pattamatta, S.K. Das, Investigation on flow maldistribution in parallel microchannel systems for integrated microelectronic cooling, *IEEE Trans. Compon. Packag. Technol.* 4 (2014) 438–450.
- [26] Y. Mishan, A. Mosyak, E. Pogrebnyak, G. Hetsroni, Effect of developing flow and thermal regime on momentum and heat transfer in micro-scale heat sink, *Int. J. Heat Mass Transfer* 50 (2007) 3100–3114.
- [27] C. Anbumeenakshi, M.R. Thansekhar, Experimental investigation of header shape and inlet configuration on flow maldistribution in microchannel, *Exp. Therm. Fluid Sci.* 75 (2016) 156–161.
- [28] G.D. Xia, J. Jiang, J. Wang, Y.L. Zhai, D.D. Ma, Effects of different geometric structures on fluid flow and heat transfer performance in microchannel heat sinks, *Int. J. Heat Mass Transfer* 80 (2015) 439–447.
- [29] C.C. Wang, K.S. Yang, J.S. Tsai, Y. Chen, Characteristics of flow distribution in compact parallel flow heat exchangers, part II: modified inlet header, *Appl. Therm. Eng.* 31 (2011) 3235–3242.
- [30] X. Liu, J. Yu, Numerical study on performances of mini-channel heat sinks with non-uniform inlets, *Appl. Therm. Eng.* 93 (2016) 856–864.
- [31] M. Mohammadi, G.N. Jovanovic, K.V. Sharp, Numerical study on flow uniformity and pressure characteristics within a microchannel array with triangular manifolds, *Comp. Chem. Eng.* 52 (2013) 134–144.
- [32] G. Kumaraguruparan, R.M. Kumaran, T. Sornakumar, T. Sundararajan, A numerical and experimental investigation of flow maldistribution in a micro-channel heat sink, *Int. Commun. Heat Mass Transfer* 38 (2011) 1349–1353.
- [33] S.G. Kandlikar, W.J. Grande, Evolution of microchannel flow passages: thermo-hydraulic performance and fabrication technology, *Heat Transf. Eng.* 24 (2003) 3–17.
- [34] S.G. Kandlikar, S. Garimella, D. Li, S. Colin, M.R. King, Chapter 3 – Single-phase liquid flow in minichannels and microchannels, *Heat Transfer and Fluid Flow in Minichannels and Microchannels*, 2006, pp. 87–136.
- [35] ANSYS Inc., *ANSYS Fluent Theory Guide*, Canonsburg, PA: release 18.0, 2016.
- [36] B. Kader, Temperature and concentration profiles in fully turbulent boundary layers, *Int. J. Heat Mass Transfer* 24 (1981) 1541–1544.
- [37] J.P. Van Doormaal, G.D. Raithby, Enhancements of the SIMPLE method for predicting incompressible fluid flows, *Num. Heat Transfer* 7 (1984) 147–163.
- [38] E.W. Lemmon, M.L. Huber, M.O. McLinden, *Reference Fluid Thermodynamic and Transport Properties – REFPROP version 9.1*, NIST, MD, 2013.
- [39] D. Copeland, Manifold microchannel heat sinks: analysis and optimization, *ASME/JSM E Therm. Eng.* 4 (1995) 169–174.
- [40] R.K. Shah, A.L. London, *Laminar Flow Forced Convection in Ducts: A Source Book for Compact Heat Exchanger Analytical Data*, Academic Press, New York, NY, 1978 (Suppl. 1).
- [41] B.J. Jones, P.S. Lee, S.V. Garimella, Infrared micro-particle image velocimetry measurements and predictions of flow distribution in a microchannel heat sink, *Int. J. Heat Mass Transfer* 51 (2008) 1877–1887.
- [42] F. Huchet, J. Comiti, P. Legentilhomme, C. Sollicec, J. Legrand, A. Montillet, Multi-scale analysis of hydrodynamics inside a network of crossing minichannels using electrodiffusion method and PIV measurements, *Int. J. Heat Fluid Flow* 29 (2008) 1411–1421.